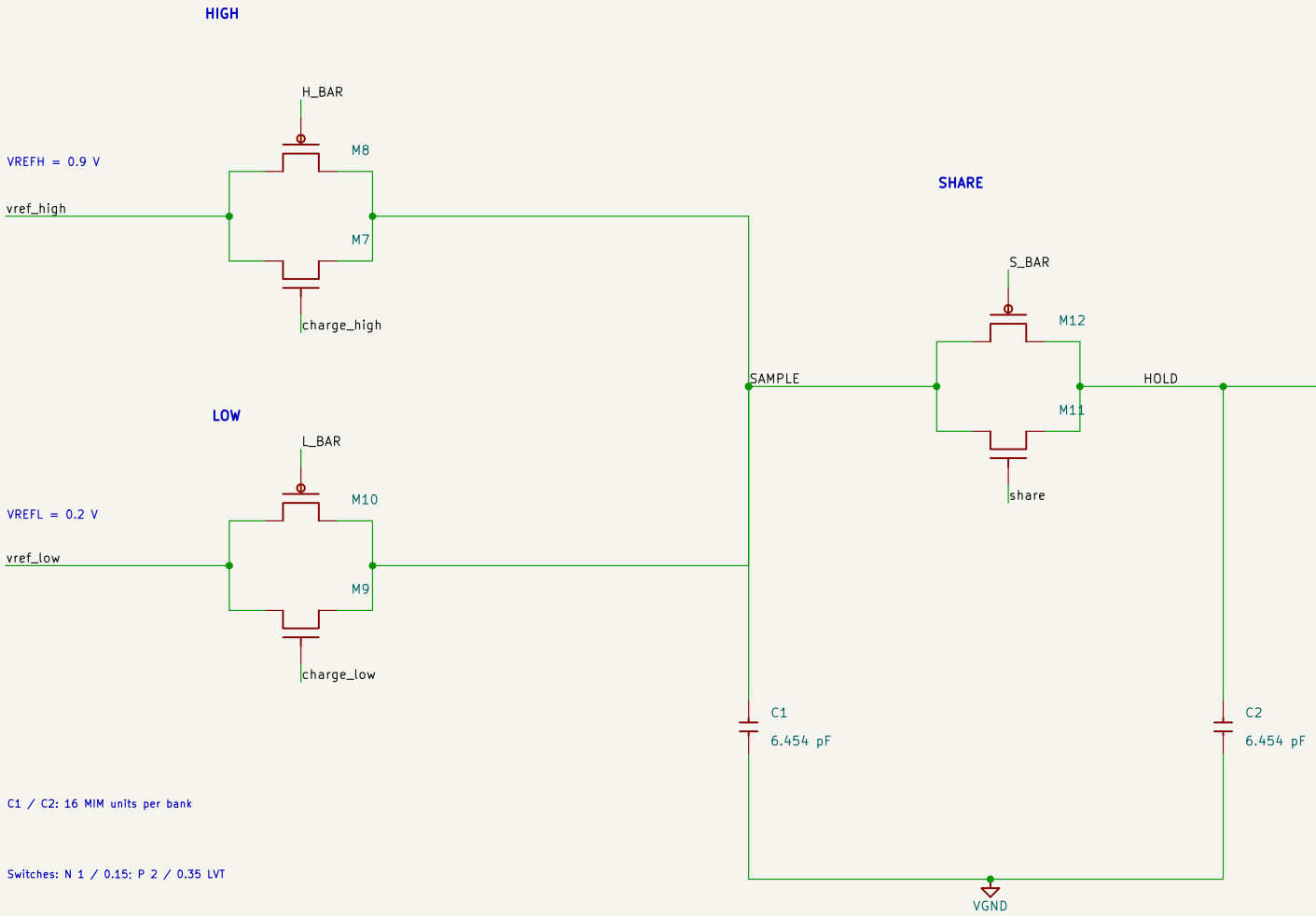


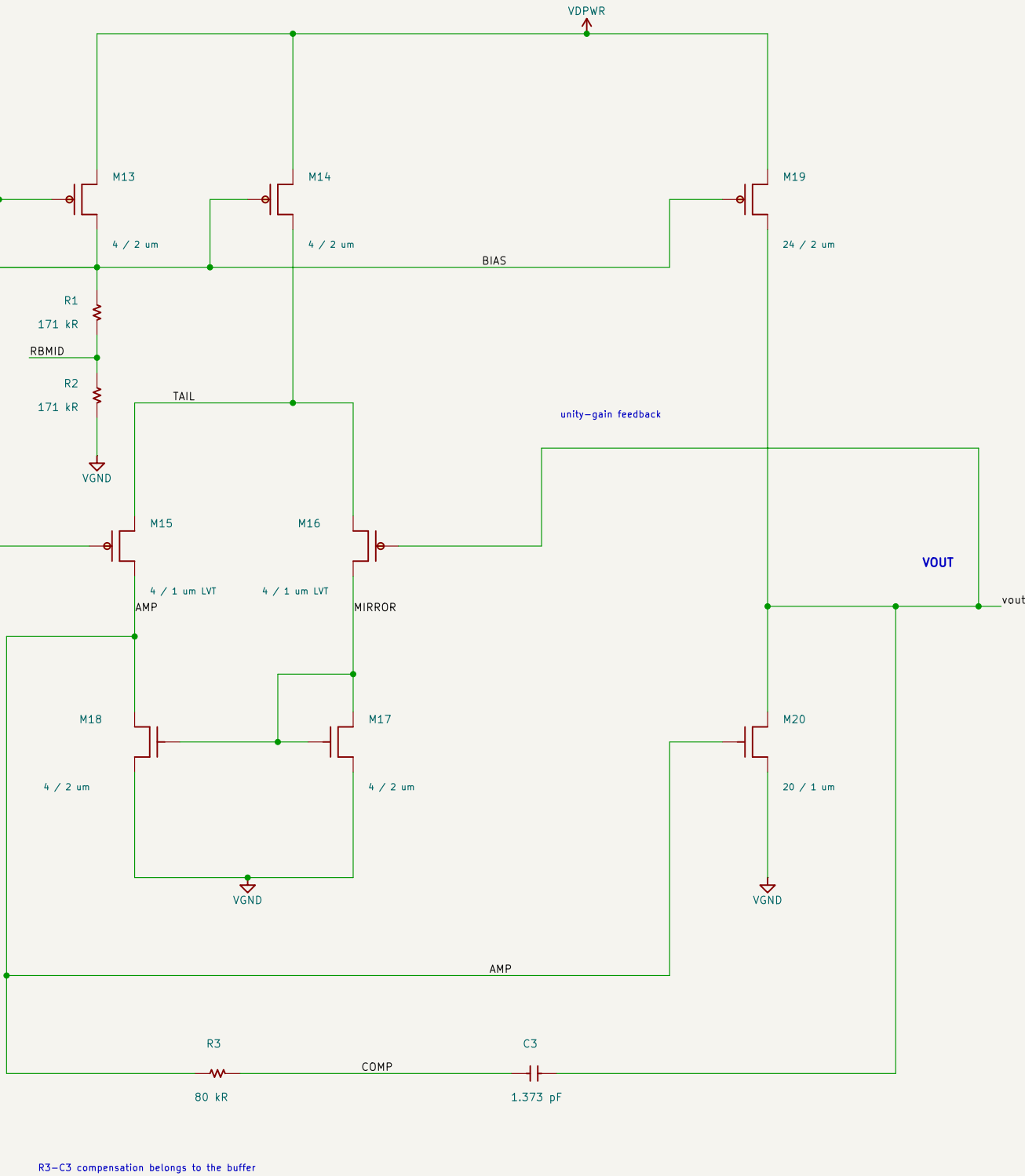
SUAREZ SERIAL CHARGE-SHARING DAC

SKY130 | 1.8 V | 8 bits, LSB first | 180 x 115 um | W / L dimensions in um

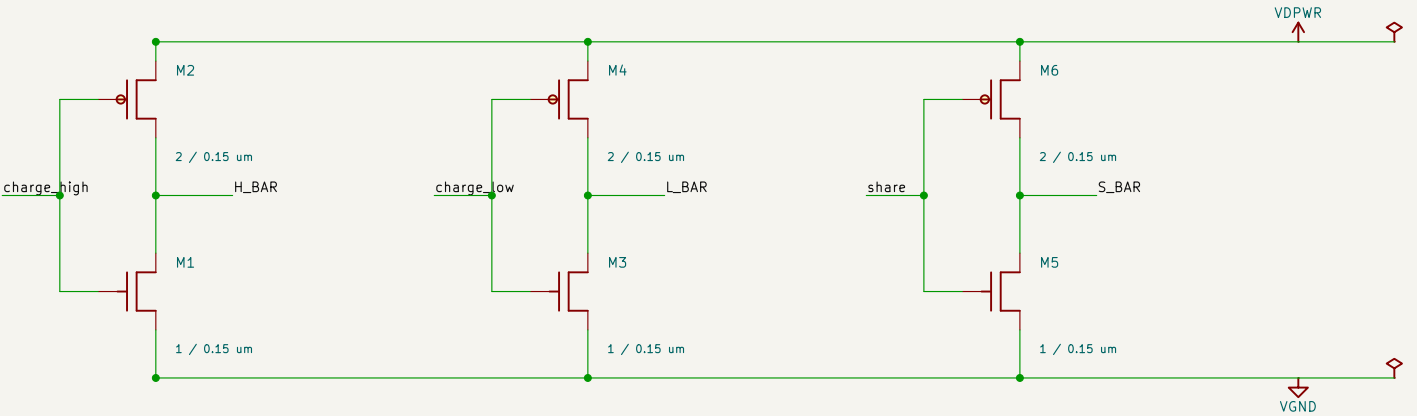
01 REFERENCE SELECTION AND CHARGE SHARING



02 UNITY-GAIN BUFFER / BIAS, FEEDBACK AND COMPENSATION



03 COMPLEMENTARY PHASE DRIVERS



OPERATING SEQUENCE

Initialize: HIGH=0, LOW=1, SHARE=1 for 8 us. Then all phases off for 0.2 us.

8 bits, LSB first: charge 2 us; dead 0.2 us; share 2 us; dead 0.2 us. Read 3 us after bit 7.

HIGH / LOW never overlap. LOW / SHARE overlap only during initialization. External controller supplies dead time.

Bodies: all NMOS -> VGND; all PMOS -> VDPWR. Body ties are explicit in SPICE / silicon.

SKY130 / Siliccluster v3

Sheet: /
File: suarez_dac.kicad_sch

Title: Suarez two-capacitor DAC

Size: A3

Date:

Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 1/1